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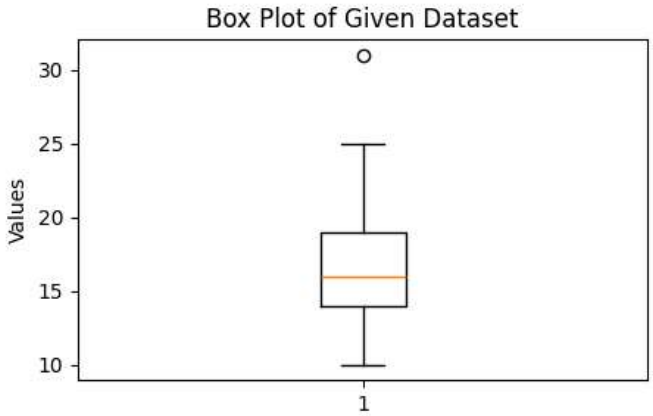
```
import numpy as np
import matplotlib.pyplot as plt

# Q1 of Part-B
data = [17, 15, 17, 20, 13, 15, 15, 16, 16, 15, 19, 12, 19,
        14, 11, 14, 16, 10, 19, 18, 20, 14, 17, 19, 16, 22,
        21, 23, 14, 12, 18, 13, 12, 25, 14, 15, 31, 17, 10, 21]

# Convert to numpy array
data = np.array(data)

# Calculate five-point summary
minimum = np.min(data)
q1 = np.percentile(data, 25)
median = np.percentile(data, 50)
q3 = np.percentile(data, 75)
maximum = np.max(data)

# Printing values
print("Five-Point Summary:")
print("Minimum = ", minimum)
print("Q1 (25th percentile)= ", q1)
print("Q2 (50th %ile/ Median) = ", median)
print("Q3 (75th percentile) = ", q3)
print("Maximum = ", maximum)
```

```
Five-Point Summary:
Minimum =  10
Q1 (25th percentile)=  14.0
Q2 (50th %ile/ Median) =  16.0
Q3 (75th percentile) =  19.0
Maximum =  31
```

```
# Identify outliers based on IQR method
iqr = q3 - q1
print(f"Inter Quartile Range (IQR): {iqr}")

lower_limit = q1 - 1.5 * iqr
upper_limit = q3 + 1.5 * iqr

print(f"Lower Bound for Outliers: {lower_limit}")
print(f"Upper Bound for Outliers: {upper_limit}")

# Outliers identified
outliers = data[(data < lower_limit) | (data > upper_limit)]
print(f"Outliers identified: {outliers}")
```

```
Inter Quartile Range (IQR): 5.0
Lower Bound for Outliers: 6.5
Upper Bound for Outliers: 26.5
Outliers identified: [31]
```

```
# Box plot
plt.figure(figsize=(5,3))
plt.boxplot(data)
plt.title('Box Plot of Given Dataset')
plt.ylabel('Values')
plt.show()
```




```
# Q2 of Part-B
import pandas as pd
data = pd.read_csv('tips.csv')
data.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with data New interactive sheet

```
# Select the required column
values = data["total_bill"]

# Compute Quartiles
Q1 = values.quantile(0.25)
Q2 = values.quantile(0.50) # Median
Q3 = values.quantile(0.75)

# Compute Decile ad Percentile
D7 = values.quantile(0.70)
P55 = values.quantile(0.55)

# Display Results
print("Partition Values for total_bills")
print("Q1 (First Quartile = )", Q1)
print("Q2 (Median = )", Q2)
print("Q3 (Third Quartile = )", Q3)
print("D7 (7th Decile = )", D7)
print("P55 (55th Percentile = )", P55)
```

```
Partition Values for total_bills
Q1 (First Quartile = ) 13.3475
Q2 (Median = ) 17.795
Q3 (Third Quartile = ) 24.127499999999998
D7 (7th Decile = ) 22.508
P55 (55th Percentile = ) 18.329
```

```
tip_values = data["tip"]

# Compute five-point summary
minimum = tip_values.min()
q1 = tip_values.quantile(0.25)
median = tip_values.quantile(0.50)
```

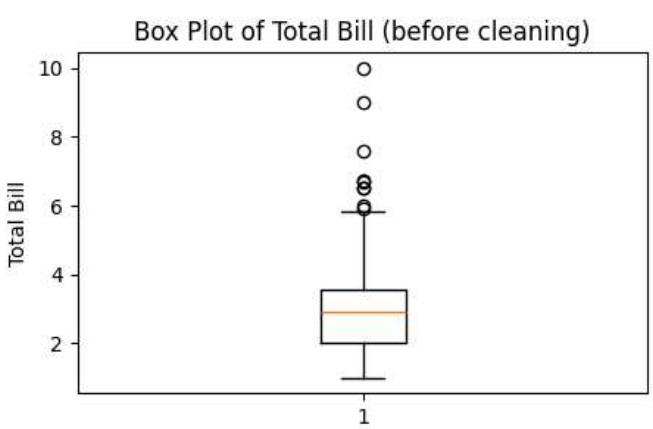
```
q3 = tip_values.quantile(0.75)
maximum = tip_values.max()

# Printing values
print("Five-Point Summary:")
print("Minimum = ", minimum)
print("Q1 = ", q1)
print("Median = ", median)
print("Q3 = ", q3)
print("Maximum = ", maximum)
```

```
Five-Point Summary:
Minimum =  1.0
Q1 =  2.0
Median =  2.9
Q3 =  3.5625
Maximum =  10.0
```

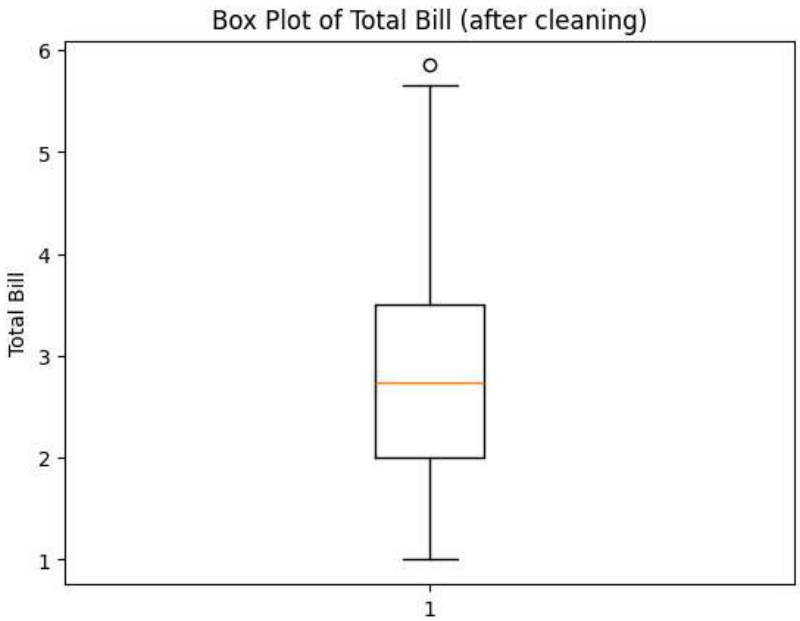
```
# Compute IQR
iqr = q3 - q1
lower_limit = q1 - 1.5 * iqr
upper_limit = q3 + 1.5 * iqr

print("IQR: ", iqr)
print("Lower Bound for Outliers: ", lower_limit)
print("Upper Bound for Outliers: ", upper_limit)

# Outliers identified
outliers = tip_values[(tip_values < lower_limit) | (tip_values > upper_limit)]
print(f"Outliers identified: {outliers}")
```

```
IQR:  1.5625
Lower Bound for Outliers:  -0.34375
Upper Bound for Outliers:  5.90625
Outliers identified: 23      7.58
47      6.00
59      6.73
141     6.70
170    10.00
183     6.50
212     9.00
214     6.50
239     5.92
Name: tip, dtype: float64
```

```
# Box plot before cleaning
import matplotlib.pyplot as plt

plt.figure(figsize=(5,3))
plt.boxplot(tip_values)
plt.title('Box Plot of Total Bill (before cleaning)')
plt.ylabel('Total Bill')
plt.show()
```



```
# Clean the data by removing outliers from 'tip' feature
cleaned_data = tip_values[(tip_values >= lower_limit) & (tip_values <= upper_limit)]

# Box plot after cleaning
plt.boxplot(cleaned_data)
plt.title('Box Plot of Total Bill (after cleaning)')
plt.ylabel('Total Bill')
plt.show()

# Display cleaned dataset size
print("Original Dataset size:", tip_values.shape)
print("Cleaned Dataset size:", cleaned_data.shape)
```




```
Original Dataset size: (244,)
Cleaned Dataset size: (235,)
```